

Day – 11 DNS server configuration Theory.

Task – 1 Set up DNS server.

- Step 1: Install BIND9 on Linux
- Open your Linux terminal and run the following commands to install the BIND name server and its utilities:
- Bash
- `sudo apt update`
- `sudo apt install bind9 bind9utils -y`
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- Step 2: Define the New Zone in BIND
- You need to tell BIND that it is now the authoritative server for musicmania.in.
- Open the local configuration file:
- Bash
- `sudo nano /etc/bind/named.conf.local`
- Add the zone definition block at the bottom of the file:
- `zone "musicmania.in" {`
- `type master;`
- `file "/etc/bind/db.musicmania.in";`
- `};`
- Save and close the file (Ctrl + O, then Enter, then Ctrl + X).
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- Step 3: Create the Zone Record File
- Next, create the actual database file that maps domain names to IP addresses for your new zone.
- Create and open the zone file:
- Bash
- `sudo nano /etc/bind/db.musicmania.in`
- Add the following records (replace 192.168.1.100 with your server or test machine's actual local IP address):
- `$TTL 86400`
- `@ IN SOA ns1.musicmania.in. admin.musicmania.in. (`
- `2026092901 ; Serial`
- `3600 ; Refresh`
- `1800 ; Retry`
- `604800 ; Expire`
- `86400 ) ; Minimum TTL`

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- ; Name Server Records
- @ IN NS ns1.musicmania.in.
- ns1 IN A 192.168.1.50
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- ; A Records for your domain and www subdomain
- @ IN A 192.168.1.100
- www IN A 192.168.1.100
- Save and close the file.
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- Step 4: Verify and Restart the DNS Server
- Check your configuration files for any syntax errors by running:
- Bash
- named-checkconf
- named-checkzone musicmania.in /etc/bind/db.musicmania.in
- Restart the BIND service to apply your changes:
- Bash
- sudo systemctl restart bind9
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- Step 5: Test Your New DNS Zone
- You can verify your new server works by querying it locally using nslookup:
- Bash
- nslookup www.musicmania.in 127.0.0.1

Task – 2 Add the protocol in DNS server.

- Step 1: Update the Zone File
- Open your zone file with root privileges:
- Bash
- sudo nano /etc/bind/db.musicmania.in
- Update the A record for www to point to 192.168.1.50, and increment the Serial number (e.g., from 2026092901 to 2026092902) so secondary servers or resolvers know the record has changed:
- \$TTL 86400
- @ IN SOA ns1.musicmania.in. admin.musicmania.in. (
- 2026092902 ; Serial (Incremented)
- 3600 ; Refresh
- 1800 ; Retry
- 604800 ; Expire

- 86400) ; Minimum TTL
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- ; Name Server Records
- @ IN NS ns1.musicmania.in.
- ns1 IN A 192.168.1.50
-
- ; Updated A Record for www subdomain
- @ IN A 192.168.1.50
- www IN A 192.168.1.50
- Save and close the file.
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- Step 2: Reload the DNS Server
- Instead of a full restart, you can gracefully reload the BIND service to apply the updated zone file:
- Bash
- sudo rndc reload
- *(Or use sudo systemctl restart bind9 if rndc is not configured).*
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- Step 3: Verify the Mapping with dig or nslookup
- Test the newly updated mapping locally by querying your BIND server directly:
- Using dig:
- Bash
- dig @127.0.0.1 www.musicmania.in +short
- Expected Output:
- 192.168.1.50
- Using nslookup:
- Bash
- nslookup www.musicmania.in 127.0.0.1
- Expected Output:
- Server: 127.0.0.1
- Address: 127.0.0.1#53
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- Name: www.musicmania.in
- Address: 192.168.1.50
- The successful return of 192.168.1.50 confirms that your local DNS server has successfully registered the updated record and is correctly resolving requests for www.musicmania.in.

Task – 3 Simulate by adding CNAME record.

- Step 1: Add the CNAME Record to the Zone File
- Open your DNS zone file:
- Bash
- `sudo nano /etc/bind/db.musicmania.in`
- Add the new CNAME record for play and increment the Serial number (e.g., from 2026092902 to 2026092903):
- `$TTL 86400`
- `@ IN SOA ns1.musicmania.in. admin.musicmania.in. (`
- `2026092903 ; Serial (Incremented)`
- `3600 ; Refresh`
- `1800 ; Retry`
- `604800 ; Expire`
- `86400 ) ; Minimum TTL`
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- `; Name Server Records`
- `@ IN NS ns1.musicmania.in.`
- `ns1 IN A 192.168.1.50`
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- `; Existing A Records`
- `@ IN A 192.168.1.50`
- `www IN A 192.168.1.50`
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- `; New CNAME Record for play subdomain`
- `play IN CNAME www.musicmania.in.`
- Save and close the file.
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- Step 2: Reload Your DNS Server
- Apply the updated configuration by reloading BIND:
- Bash
- `sudo rndc reload`
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- Step 3: Test and Verify the CNAME Resolution
- Test your configuration using dig or nslookup to see how the resolver handles the alias:
- Using dig:
- Bash
- `dig @127.0.0.1 play.musicmania.in`
- Expected Output (Excerpt):
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- `;play.musicmania.in. IN A`
- `play.musicmania.in. 86400 IN CNAME www.musicmania.in.`
- `www.musicmania.in. 86400 IN A 192.168.1.50`

- Using nslookup:
- Bash
- nslookup play.musicmania.in 127.0.0.1
- Expected Output:
- Server: 127.0.0.1
- Address: 127.0.0.1#53
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- Non-authoritative answer:
- play.musicmania.in canonical name = www.musicmania.in.
- Name: www.musicmania.in
- Address: 192.168.1.50

Task – 4 Based on scenario.

- Step 1: Check Global and Local Configuration Syntax
- Before looking at individual records, verify that your main configuration files (named.conf and named.conf.local) contain no syntax errors, missing semicolons, or incorrect path declarations.
- The Command:
- Bash
- sudo named-checkconf
- What it checks: This command parses your BIND configuration files. If it returns no output, your syntax is clean. If there is a typo (like a missing ; in named.conf.local), it will output the exact file name and line number causing the failure.
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- Step 2: Validate the Zone File and Serial Numbers
- If the configuration is correct, the issue often lies inside the zone file (db.musicmania.in), such as a missing trailing dot (.) on a Fully Qualified Domain Name (FQDN) or an un-incremented serial number.
- The Command:
- Bash
- sudo named-checkzone musicmania.in /etc/bind/db.musicmania.in
- What it checks: This verifies that all records (A, NS, CNAME) are syntactically valid and that BIND can successfully load the zone database. Look for an OK response; if it fails, it will point directly to the problematic record line.
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- Step 3: Inspect Service Health and System Logs
- If BIND refuses to answer queries, the daemon might have encountered a fatal error during startup and stopped running.

- The Commands:
- Bash
- `sudo systemctl status bind9`
- `journalctl -u bind9 -n 50 --no-pager`
- What they check: `systemctl status` tells you if the service is actively running or failed. Checking the journal logs (`journalctl`) reveals runtime errors, such as permission denied issues on the zone files or failure to bind to port 53.
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- Step 4: Apply Fixes, Reload, and Test
- Once you correct any identified issues in your configuration or zone files, reload the DNS service and verify functionality using a targeted query:
- Reload BIND gracefully:
- Bash
- `sudo rndc reload`
- Test direct resolution using `dig`:
- Bash
- `dig @127.0.0.1 www.musicmania.in +short`
- If it returns your target IP (e.g., 192.168.1.50), your troubleshooting was successful and resolution is restored!
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